The University of Azad Jammu and Kashmir

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Course Title: Database System

Course Code: CS-2204

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LAB 11

1. Introduction

Scalar functions operate on one row at a time and return one value for each row. In this lab, string functions are used to clean, format, extract, and modify text data stored in the Customer and Product tables.

The main string functions used in this lab are:

- TRIM()
- UPPER()
- LOWER()
- CHAR_LENGTH()
- CONCAT()
- SUBSTRING()
- LEFT()
- REPLACE()
- LPAD()
- LOCATE()

Create Database

2. Create and Insert Table (Customer , Product)

Table	Action	Rows	Type	Collation	Size	Overhead
customer	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
product	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
2 tables	Sum	0	InnoDB	utf8mb4_general_ci	32.0 KiB	0 B

```
SELECT * FROM `customer`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	CustID	CustName	Email	City	Phone	JoinDate	DOB
Edit Copy Delete	1	Ali Khan	ali.khan@MAIL.com	Lahore	0300-1112233	2022-01-15	1995-04-12
Edit Copy Delete	2	Sara Iqbal	sara@example.com	Karachi	0301-4445566	2022-04-22	1998-11-20
Edit Copy Delete	3	HAMZA RAZA	hamza@example.com	Lahore	0302-7778899	2023-02-10	1997-08-05
Edit Copy Delete	4	Ayesha Noor	NULL	Islamabad	0303-1234567	2023-05-18	1999-02-14
Edit Copy Delete	5	bilal ahmed	bilal@MAIL.COM	Karachi	0304-2345678	2023-09-01	2000-06-30
Edit Copy Delete	6	Fatima Sheikh	fatima@example.com	NULL	0305-3456789	2024-01-12	1996-10-25
Edit Copy Delete	7	Usman Tariq	usman@example.com	Lahore	NULL	2024-06-30	2001-03-18
Edit Copy Delete	8	Maira Javed	maira@example.com	Islamabad	0307-5678901	2024-08-25	1994-12-09

```
SELECT * FROM `product`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	ProdID	ProdName	Category	Price	StockQty	LaunchDate
Edit Copy Delete	101	Laptop Pro 15	Electronics	185000.00	12	2023-03-10
Edit Copy Delete	102	Wireless Mouse	Electronics	2500.00	50	2022-07-22
Edit Copy Delete	103	USB-C Cable	Electronics	800.00	100	2021-11-05
Edit Copy Delete	104	Office Chair	Furniture	18500.00	8	2023-01-15
Edit Copy Delete	105	Standing Desk	Furniture	45000.50	5	2024-02-28
Edit Copy Delete	106	Notebook A4	Stationery	350.00	200	2020-04-01
Edit Copy Delete	107	Ballpoint Pen 10pk	Stationery	450.00	150	2020-04-01
Edit Copy Delete	108	Coffee Beans 1kg	Grocery	1899.99	30	2023-09-20
Edit Copy Delete	109	Green Tea Box	Grocery	650.00	45	2022-12-12
Edit Copy Delete	110	Bluetooth Speaker	Electronics	7500.00	18	2024-05-18

3.PART B — NUMERIC AND DATE/TIME FUNCTIONS

In Part B, numeric and date/time scalar functions are used to perform calculations, rounding, date extraction, date formatting, date differences, and date arithmetic. These functions transform raw database values into meaningful and human-readable information.

B1 — Apply a 15% Discount

Objective:

Apply a 15% discount to every product and display the original price and discounted price rounded to two decimal places.

SQL Query:

```
SELECT
   ProdName,
   Price,
   ROUND(Price * 0.85, 2)
AS
DiscountedPrice
FROM Product;
```

Functions Used: ROUND()

Explanation: A 15% discount means the customer pays 85% of the original price. ROUND() is used to round the result to two decimal places. **Formula:** Discounted Price = Price × 0.85

```
SELECT ProdName, Price, ROUND(Price * 0.85, 2) AS DiscountedPrice FROM Product;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key:

Extra options

←T→	ProdName	Price	DiscountedPrice
Edit Copy Delete	Laptop Pro 15	185000.00	157250.00
Edit Copy Delete	Wireless Mouse	2500.00	2125.00
Edit Copy Delete	USB-C Cable	800.00	680.00
Edit Copy Delete	Office Chair	18500.00	15725.00
Edit Copy Delete	Standing Desk	45000.50	38250.43
Edit Copy Delete	Notebook A4	350.00	297.50
Edit Copy Delete	Ballpoint Pen 10pk	450.00	382.50
Edit Copy Delete	Coffee Beans 1kg	1899.99	1614.99
Edit Copy Delete	Green Tea Box	650.00	552.50
Edit Copy Delete	Bluetooth Speaker	7500.00	6375.00

Console

B2 —Calculate 17% Sales Tax

Objective:

Calculate 17% sales tax for every product and display the final price including tax.

SQL Query:

```
SELECT
ProdName,
ROUND(Price * 0.17, 2) AS
Tax,
ROUND(Price + (Price * 0.17), 2) AS PriceWithTax
FROM Product;
```

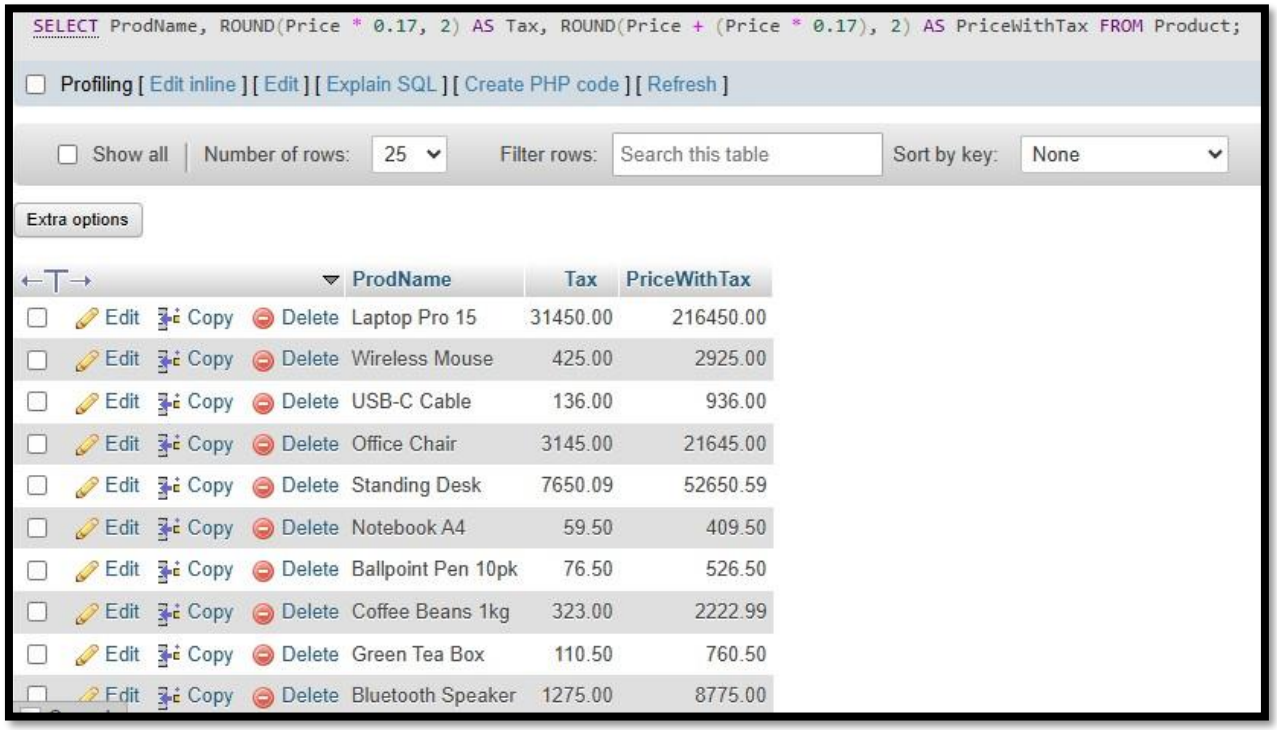
Functions Used: ROUND()

Explanation: The tax is calculated by multiplying the product price by 17%. The tax is then added to the original price.

Formula:

Tax = Price × 0.17

Price With Tax = Price + Tax



```
SELECT ProdName, ROUND(Price * 0.17, 2) AS Tax, ROUND(Price + (Price * 0.17), 2) AS PriceWithTax FROM Product;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

			ProdName	Tax	PriceWithTax
Edit	Copy	Delete	Laptop Pro 15	31450.00	216450.00
Edit	Copy	Delete	Wireless Mouse	425.00	2925.00
Edit	Copy	Delete	USB-C Cable	136.00	936.00
Edit	Copy	Delete	Office Chair	3145.00	21645.00
Edit	Copy	Delete	Standing Desk	7650.09	52650.59
Edit	Copy	Delete	Notebook A4	59.50	409.50
Edit	Copy	Delete	Ballpoint Pen 10pk	76.50	526.50
Edit	Copy	Delete	Coffee Beans 1kg	323.00	2222.99
Edit	Copy	Delete	Green Tea Box	110.50	760.50
Edit	Copy	Delete	Bluetooth Speaker	1275.00	8775.00

B3 — Floor and Ceiling of Price Divided by 1000

Objective:

Calculate the floor and ceiling values of each product's price divided by 1000.

SQL Query:

```
SELECT
  ProdName,
  Price,
  FLOOR(Price / 1000) AS
FloorVal,
  CEIL(Price / 1000) AS
CeilVal
```

FROM Product;

Functions Used: FLOOR(), CEIL()

Explanation: FLOOR() returns the largest integer less than or equal to the value.

CEIL() returns the smallest integer greater than or equal to the value.

```
SELECT ProdName, Price, FLOOR(Price / 1000) AS FloorVal, CEIL(Price / 1000) AS CeilVal FROM Product;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→			ProdName	Price	FloorVal	CeilVal
Edit	Copy	Delete	Laptop Pro 15	185000.00	185	185
Edit	Copy	Delete	Wireless Mouse	2500.00	2	3
Edit	Copy	Delete	USB-C Cable	800.00	0	1
Edit	Copy	Delete	Office Chair	18500.00	18	19
Edit	Copy	Delete	Standing Desk	45000.50	45	46
Edit	Copy	Delete	Notebook A4	350.00	0	1
Edit	Copy	Delete	Ballpoint Pen 10pk	450.00	0	1
Edit	Copy	Delete	Coffee Beans 1kg	1899.99	1	2
Edit	Copy	Delete	Green Tea Box	650.00	0	1
Edit	Copy	Delete	Bluetooth Speaker	7500.00	7	8

B4 — Round Price to the Nearest Hundred

Objective:

Round each product's price to the nearest hundred.

SQL Query:

SELECT

ProdName,

Price,

ROUND(Price, -2) AS

RoundedPrice

FROM Product;

Function Used: ROUND()

Explanation: The second argument -2 tells MySQL to round the number to the nearest hundred.

Example:

185000 → 185000

18550 → 18600

Showing rows 0 - 9 (10 total, Query took 0.0005 seconds.)

```
SELECT ProdName, Price, ROUND(Price, -2) AS RoundedPrice FROM Product;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by ke

Extra options

←T→	ProdName	Price	RoundedPrice
Edit Copy Delete	Laptop Pro 15	185000.00	185000
Edit Copy Delete	Wireless Mouse	2500.00	2500
Edit Copy Delete	USB-C Cable	800.00	800
Edit Copy Delete	Office Chair	18500.00	18500
Edit Copy Delete	Standing Desk	45000.50	45000
Edit Copy Delete	Notebook A4	350.00	400
Edit Copy Delete	Ballpoint Pen 10pk	450.00	500
Edit Copy Delete	Coffee Beans 1kg	1899.99	1900
Edit Copy Delete	Green Tea Box	650.00	700
Edit Copy Delete	Bluetooth Speaker	7500.00	7500

Console

B5 — Find Products with Odd Product IDs

Objective:

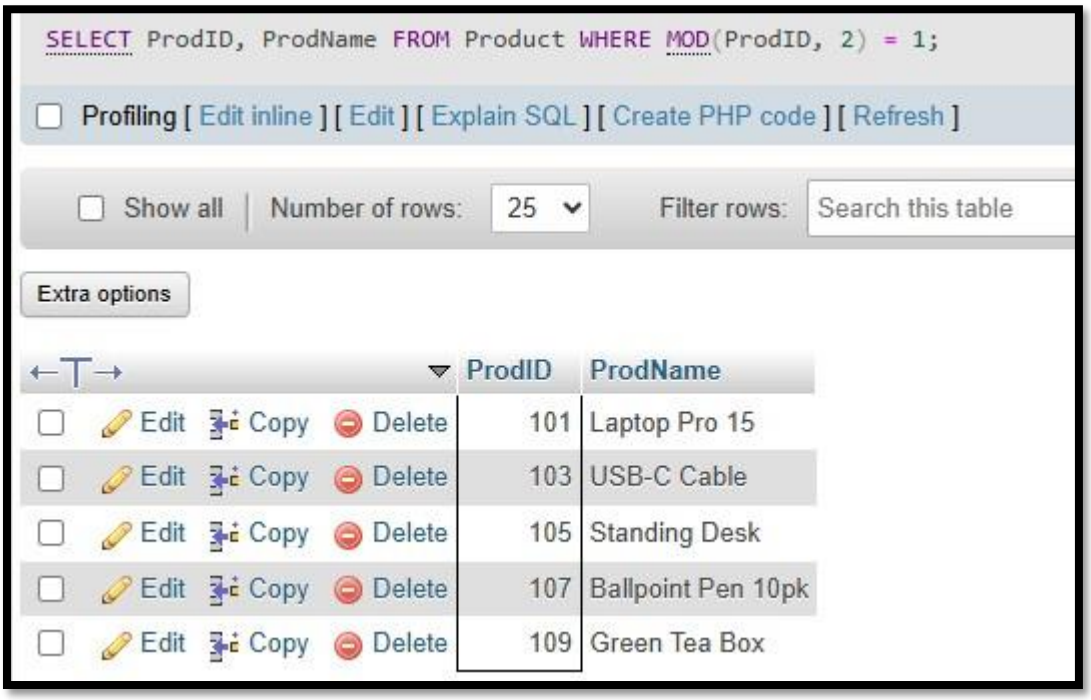
List all products whose ProdID is an odd number.

SQL Query:

```
SELECT
    ProdID,
    ProdName
FROM Product
  WHERE MOD(ProdID, 2) = 1;
```

Function Used: MOD()

Explanation: MOD() returns the remainder after division. If the remainder after dividing ProdID by 2 is 1, the number is odd.



B6 — Extract Year, Month Name and Day of Week

Objective:

Display the year, month name, and day of the week for each customer's joining date.

SQL Query:

```
SELECT
    CustID,
    CustName,
    YEAR(JoinDate) AS JoinYear,
    MONTHNAME(JoinDate) AS JoinMonth,
    DAYNAME(JoinDate) AS JoinDay
FROM Customer;
```

Functions Used: YEAR(), MONTHNAME(), DAYNAME()

Explanation:

YEAR() extracts the year.

MONTHNAME() returns the full month name.

DAYNAME() returns the name of the day.

```
SELECT CustID, CustName, YEAR(JoinDate) AS JoinYear, MONTHNAME(JoinDate) AS JoinMonth, DAYNAME(JoinDate) AS JoinDay FROM Customer;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→	CustID	CustName	JoinYear	JoinMonth	JoinDay
Edit Copy Delete	1	Ali Khan	2022	January	Saturday
Edit Copy Delete	2	Sara Iqbal	2022	April	Friday
Edit Copy Delete	3	HAMZA RAZA	2023	February	Friday
Edit Copy Delete	4	Ayesha Noor	2023	May	Thursday
Edit Copy Delete	5	bilal ahmed	2023	September	Friday
Edit Copy Delete	6	Fatima Sheikh	2024	January	Friday
Edit Copy Delete	7	Usman Tariq	2024	June	Sunday
Edit Copy Delete	8	Maira Javed	2024	August	Sunday

B7—Format Date of Birth

Objective:

Display each customer's date of birth in the format DD-Month-YYYY.

SQL Query:

```
SELECT
CustName,
DOB,
DATE_FORMAT(DOB,
'%d-%M-%Y') AS FormattedDOB
FROM Customer;
```

Function Used: DATE_FORMAT()

Explanation: DATE_FORMAT() changes a date into a readable format.

For example:

1995-04-12 ***becomes:***

12-April-1995

```
SELECT CustName, DOB, DATE_FORMAT(DOB, '%d-%M-%Y') AS FormattedDOB FROM Customer;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→	CustName	DOB	FormattedDOB
Edit Copy Delete	Ali Khan	1995-04-12	12-April-1995
Edit Copy Delete	Sara Iqbal	1998-11-20	20-November-1998
Edit Copy Delete	HAMZA RAZA	1997-08-05	05-August-1997
Edit Copy Delete	Ayesha Noor	1999-02-14	14-February-1999
Edit Copy Delete	bilal ahmed	2000-06-30	30-June-2000
Edit Copy Delete	Fatima Sheikh	1996-10-25	25-October-1996
Edit Copy Delete	Usman Tariq	2001-03-18	18-March-2001
Edit Copy Delete	Maira Javed	1994-12-09	09-December-1994

B8—Calculate Current Age

Objective:

Calculate the current age of every customer in years.

SQL Query:

```
SELECT
    CustName,
    DOB,
        TIMESTAMPDIFF(YEAR,
    DOB, CURDATE()) AS Age
FROM Customer;
```

Functions Used: TIMESTAMPDIFF(), CURDATE()

Explanation:

TIMESTAMPDIFF() calculates the difference between the date of birth and today's date in years.

CURDATE() returns the current date.

```
SELECT CustName, DOB, TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age FROM Customer;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort b

Extra options

←T→			CustName	DOB	Age
Edit	Copy	Delete	Ali Khan	1995-04-12	31
Edit	Copy	Delete	Sara Iqbal	1998-11-20	27
Edit	Copy	Delete	HAMZA RAZA	1997-08-05	29
Edit	Copy	Delete	Ayesha Noor	1999-02-14	27
Edit	Copy	Delete	bilal ahmed	2000-06-30	26
Edit	Copy	Delete	Fatima Sheikh	1996-10-25	29
Edit	Copy	Delete	Usman Tariq	2001-03-18	25
Edit	Copy	Delete	Maira Javed	1994-12-09	31

B9 — Calculate Days Since Joining

Objective:

Calculate how many days have passed since each customer joined.

SQL Query:

```
SELECT
CustName,
JoinDate,
DATEDIFF(CURDATE(),
JoinDate) AS DaysSinceJoin
FROM Customer;
```

Functions Used: DATEDIFF(), CURDATE()

Explanation: DATEDIFF() calculates the difference between two dates in days.

The calculation is:

Today's Date - Join Date

```
SELECT CustName, JoinDate, DATEDIFF(CURDATE(), JoinDate) AS DaysSinceJoin FROM Customer;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: No

Extra options

			CustName	JoinDate	DaysSinceJoin
Edit	Copy	Delete	Ali Khan	2022-01-15	1707
Edit	Copy	Delete	Sara Iqbal	2022-04-22	1610
Edit	Copy	Delete	HAMZA RAZA	2023-02-10	1316
Edit	Copy	Delete	Ayesha Noor	2023-05-18	1219
Edit	Copy	Delete	bilal ahmed	2023-09-01	1113
Edit	Copy	Delete	Fatima Sheikh	2024-01-12	980
Edit	Copy	Delete	Usman Tariq	2024-06-30	810
Edit	Copy	Delete	Maira Javed	2024-08-25	754

B10 — Customers Who Joined in 2023

Objective:

List customers who joined during the year 2023 using a date function.

SQL Query:

```
SELECT
CustID,
CustName,
JoinDate
FROM Customer
```

```
WHERE YEAR(JoinDate) =
                    2023;
```

Function Used: YEAR()

Explanation: YEAR() extracts the year from JoinDate. The query then filters customers whose joining year is 2023.

B11 — Products Launched in Q4

Objective:

List products launched during October, November, or December of any year.

SQL Query:

```
SELECT
    ProdID,
    ProdName,
    LaunchDate
FROM Product
WHERE
  MONTH(LaunchDate) IN (10,
11, 12);
```

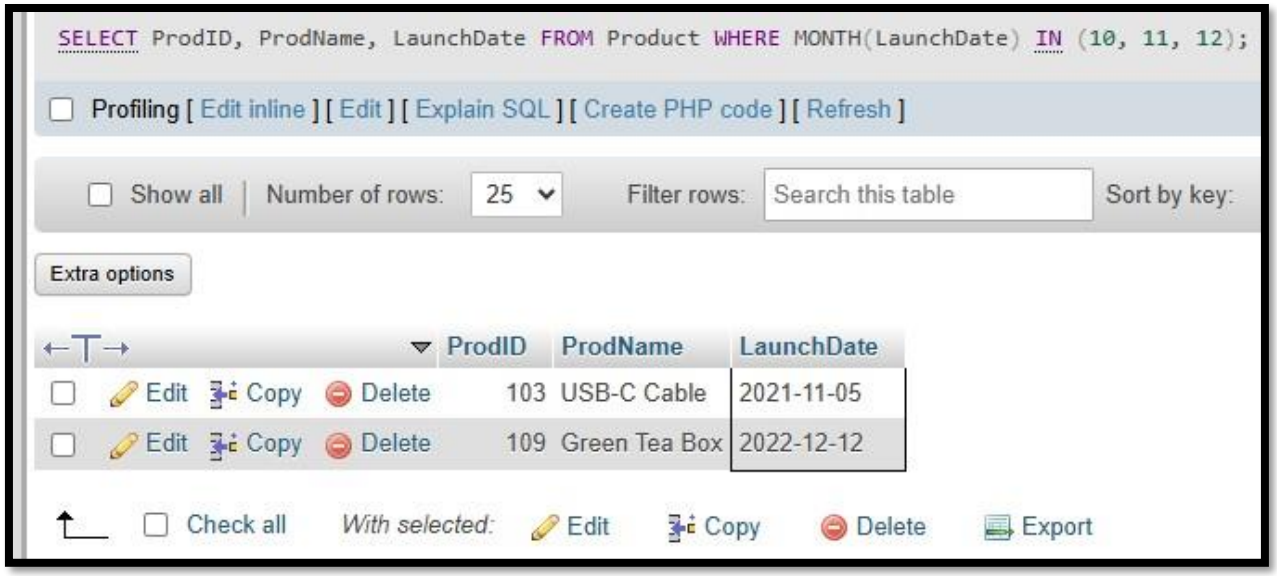
Function Used: MONTH()

Explanation: Q4 consists of three months:

October = 10

November = 11
December = 12

The IN operator checks whether the month belongs to these three values.



B12 — Customers Who Joined in the Last 6 Months

Objective:

Find customers who joined within the last six months from today's date.

SQL Query:

```
SELECT
  CustID,
  CustName,
  JoinDate
FROM Customer
```

```
WHERE JoinDate >=
DATE_SUB(CURDATE(),
INTERVAL 6 MONTH);
```

Functions Used: DATE_SUB(), CURDATE()

Explanation: DATE_SUB() subtracts six months from today's date. Customers whose JoinDate is on or after that date are selected.

B13—Product Age in Days

Objective:

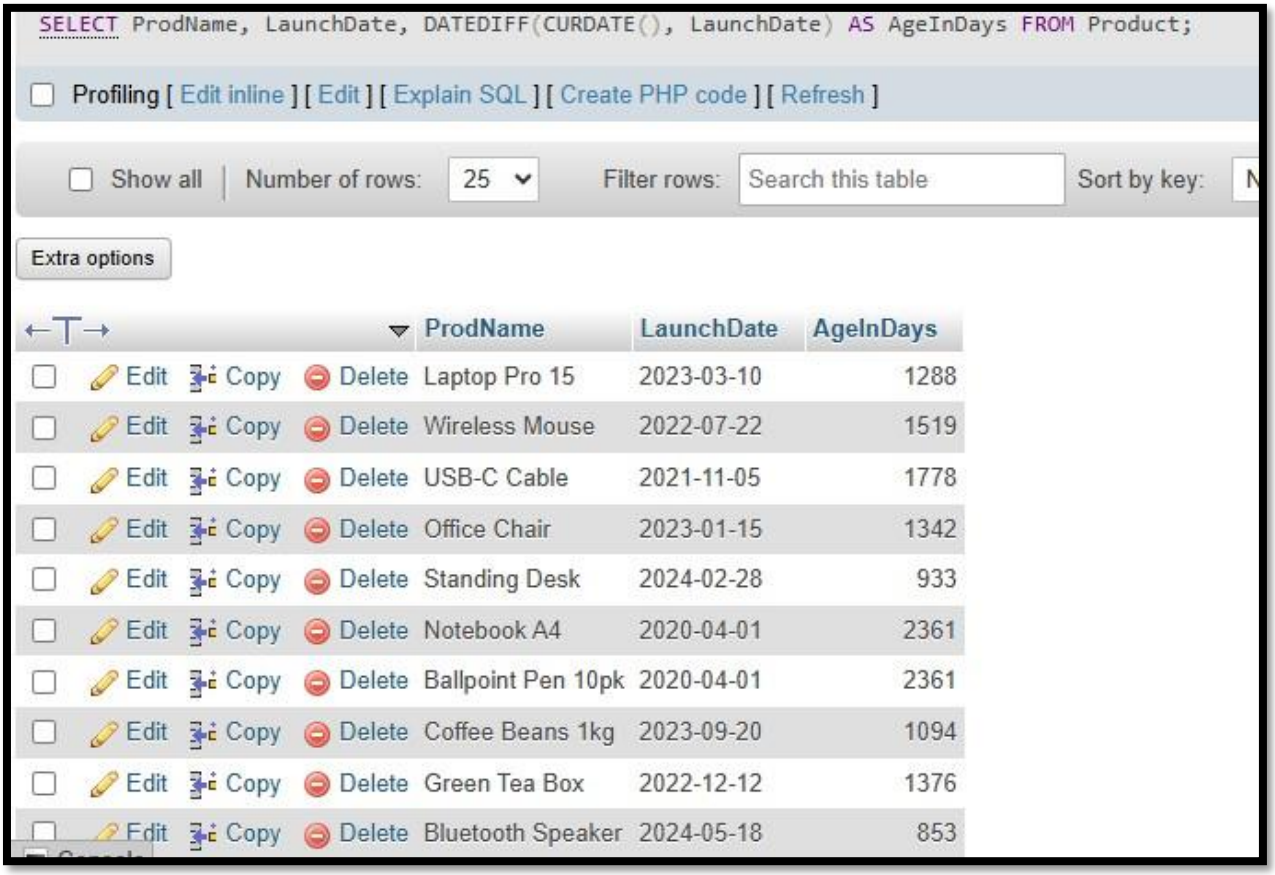
Calculate how many days have passed since each product was launched.

SQL Query:

```
SELECT
ProdName,
LlaunchDate,
DATEDIFF(CURDATE(),
LaunchDate) AS AgeInDays
FROM Product;
```

Functions Used: DATEDIFF(), CURDATE()

Explanation: The query calculates the number of days between today's date and the product's launch date.



```
SELECT ProdName, LaunchDate, DATEDIFF(CURDATE(), LaunchDate) AS AgeInDays FROM Product;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key:

Extra options

←T→			ProdName	LaunchDate	AgeInDays
Edit	Copy	Delete	Laptop Pro 15	2023-03-10	1288
Edit	Copy	Delete	Wireless Mouse	2022-07-22	1519
Edit	Copy	Delete	USB-C Cable	2021-11-05	1778
Edit	Copy	Delete	Office Chair	2023-01-15	1342
Edit	Copy	Delete	Standing Desk	2024-02-28	933
Edit	Copy	Delete	Notebook A4	2020-04-01	2361
Edit	Copy	Delete	Ballpoint Pen 10pk	2020-04-01	2361
Edit	Copy	Delete	Coffee Beans 1kg	2023-09-20	1094
Edit	Copy	Delete	Green Tea Box	2022-12-12	1376
Edit	Copy	Delete	Bluetooth Speaker	2024-05-18	853

B14 — Date 90 Days After Launch

Objective:

Calculate the date exactly 90 days after each product's launch date.

SQL Query:

```
SELECT
    ProdName,
    LaunchDate,
    DATE_ADD(LaunchDate,
    INTERVAL 90 DAY) AS
    NinetyDaysLater
FROM Product;
```

Function Used: DATE_ADD()

Explanation: DATE_ADD() adds a specified amount of time to a date. Here, 90 days are added to each product's LaunchDate.

```
SELECT ProdName, LaunchDate, DATE_ADD(LaunchDate, INTERVAL 90 DAY) AS NinetyDaysLater FROM Product;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	ProdName	LaunchDate	NinetyDaysLater
Edit Copy Delete	Laptop Pro 15	2023-03-10	2023-06-08
Edit Copy Delete	Wireless Mouse	2022-07-22	2022-10-20
Edit Copy Delete	USB-C Cable	2021-11-05	2022-02-03
Edit Copy Delete	Office Chair	2023-01-15	2023-04-15
Edit Copy Delete	Standing Desk	2024-02-28	2024-05-28
Edit Copy Delete	Notebook A4	2020-04-01	2020-06-30
Edit Copy Delete	Ballpoint Pen 10pk	2020-04-01	2020-06-30
Edit Copy Delete	Coffee Beans 1kg	2023-09-20	2023-12-19
Edit Copy Delete	Green Tea Box	2022-12-12	2023-03-12
Edit Copy Delete	Bluetooth Speaker	2024-05-18	2024-08-16

Console

B15 — Combined Customer Summary

Objective:

Create a single summary column containing the customer's uppercase name, age, and joining month/year.

SQL Query:

```
SELECT
   CONCAT(
      'Hello ',

UPPER(TRIM(CustName)),
      ', age ',

TIMESTAMPDIFF(YEAR,
DOB, CURDATE()),
      ', joined ',
```

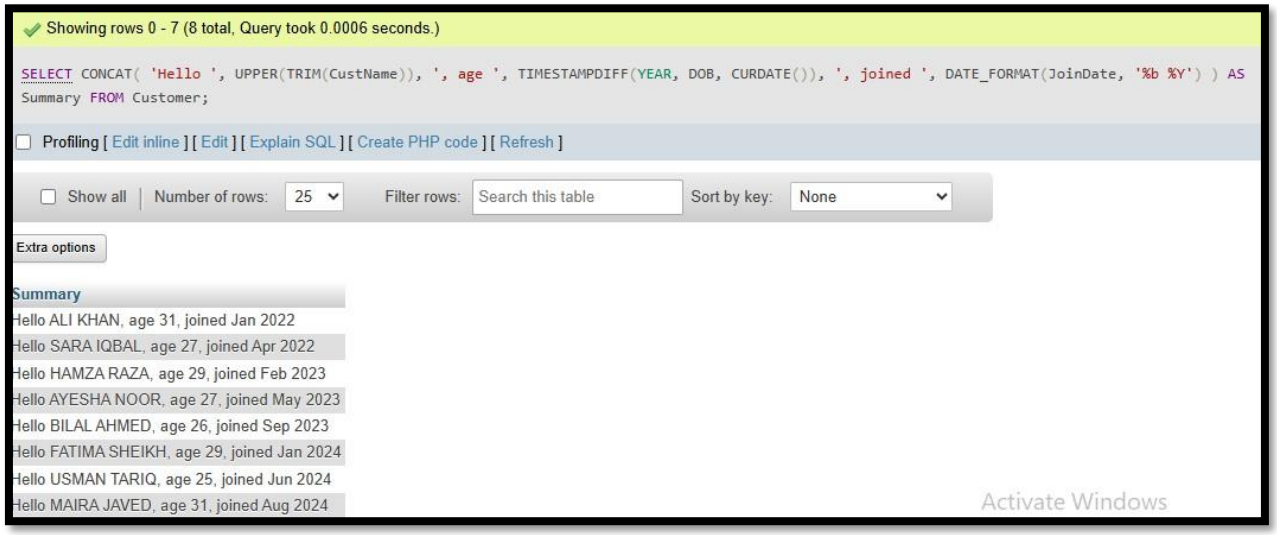
```
DATE_FORMAT(JoinDate,
'%b %Y')
   ) AS Summary
 FROM Customer;
```

Required Format: Hello ALI KHAN, age 31, joined Jan 2022

Functions Used:

- CONCAT()
- UPPER()
- TRIM()
- TIMESTAMPDIFF()
- CURDATE()
- DATE_FORMAT() **Example Output:**

Hello ALI KHAN, age 31, joined Jan 2022



Conclusion

In Part B, numeric and date/time scalar functions were used to perform calculations and transform database values into meaningful information. The tasks demonstrated discounts, sales tax, rounding, odd-number filtering, date extraction, date formatting, age calculation, date differences, date filtering, and date arithmetic.

*****THE END*****